

Final Business Proposal

Team Yellow “Clarity”

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The Virginia Division of Purchases and Supply (DPS) and the Virginia Department of Corrections (VADOC) use procurement data as their main resource for making strategic operational and compliance-based decisions. The existing reporting system experiences performance issues because it depends on manual work, uses inconsistent key performance indicator (KPI) calculations, and suffers from ongoing data quality problems, which require additional work to achieve correct results. The project documentation shows that the DPS and VADOC teams need to handle slow reporting processes, which require manual work and present problems with unclear formulas that prevent them from creating accurate and timely dashboards. The leadership team needs straightforward insights, yet the analysts must spend extra time to check data, while the operations team faces work interruptions because of obsolete reporting techniques. The organization faces three main challenges, which result in decreased efficiency, increased workload, and restricted capacity to make secure decisions based on data analysis.

To address these issues, Team Yellow developed a centralized Business Intelligence solution built in Tableau, guided by our one-word theme: Clarity. Our solution transforms raw procurement data into a unified, automated, and easy-to-use reporting system. Tableau serves as a “single pane of glass” where all users, leadership, analysts, and operations can access the same accurate information. The system contains centralized data storage together with automated KPI logic that operates through Tableau Prep and interactive dashboards, which display standardized formulas and built-in compliance tracking for SWaM and procurement policies. The off-the-shelf nature of Tableau makes it easy to use while supporting system growth and requiring only basic training for users to adopt it across multiple user groups.

Our solution produces optimal results because it combines advanced technical capabilities with a design that focuses on human needs. The system achieves its technical objectives by automating manual reporting processes while establishing standardized KPI logic, eliminating spreadsheet usage, and enabling predictive analytics of spending patterns. The system enhances operational efficiency through its ability to create a better understanding and reduce confusion while enabling employees to work together more effectively. The solution meets all RFP requirements because it exhibits both technical value and non-technical value through its demonstration of operational technology and its presentation of a practical solution that can be expanded within the semester project constraints. The value delivered is clear: faster reporting, more accurate insights, reduced manual work, and improved decision-making across the organization.

To ensure successful adoption, we applied structured change-management frameworks. The Traffic Light Model showed different levels of readiness, which showed that leadership had strong support, while analysts showed moderate hope, and operational teams showed the most resistance because they worried about their work demands, system precision, and operational equipment. The $P + R > E + R$ model implementation required us to create a positive vision while supplying training resources and documentation materials, and open data validation processes to succeed against established behaviors and their resistance. The approach creates an environment where users receive complete information while they transition to the new system with all necessary support. Our documentation explains that people who receive support from others will demonstrate a higher willingness to accept the solution.

Our architecture integrates four core components: data sources, data cleaning through Tableau Prep, Tableau dashboards, and a standardized KPI reporting layer. The system operates to provide a direct path for data movement, which starts with raw inputs and ends with complete insights. We work together with DPS and VADOC to improve data quality while we check calculation accuracy and confirm that the system operates according to actual business requirements. The technical implementation included building automated Prep flows, designing KPI logic, creating interactive dashboards, validating data with the client, and ensuring the final system is both accurate and easy to use.

The final value of our solution is grounded in clarity in data, clarity in workflows, and clarity in decision-making. Our dashboards support users with different technical backgrounds by making insights understandable rather than overwhelming. Shared clarity kept our team aligned throughout development, and it will continue to support DPS and VADOC as they adopt the system. The human outcome is simple: when reporting becomes faster, cleaner, and more transparent, users gain confidence, reduce rework, and make better decisions. Adoption will succeed because the pain of inefficient manual reporting is far greater than the effort required to learn a modern, automated system.

Each team member contributed meaningfully to this project. Bethlehem led the introduction, problem framing, and stakeholder analysis. Breanna developed the solution overview and articulated why our approach is optimal. Yabby led a change-management strategy using the Traffic Light and $P + R > E + R$ models. Yanit handled technical implementation, KPI logic, and dashboard development. Faustina delivered the live demo and explained the ease of

use and final value. Together, we operated as an agile, cooperative team focused on delivering clarity through data validation, refined insights, and sustainable BI practices.

In conclusion, Team Yellow's solution provides DPS and VADOC with a centralized, automated, and user-friendly BI system that directly addresses their reporting challenges. The integration of technical development together with human-centered change management enables us to provide customers with a comprehensive solution that they can trust and implement, and expand their usage. Our organization establishes accessibility for all users through our Clarity theme, which enables users to access precise information while decreasing their manual tasks and making secure choices. Our team functions as the ideal selection for this project because we provide the three things needed for success: a clear understanding, valuable outcomes, and a permanent solution.